Depth and Shape Inference (III)

Introduction to Computational and Biological Vision

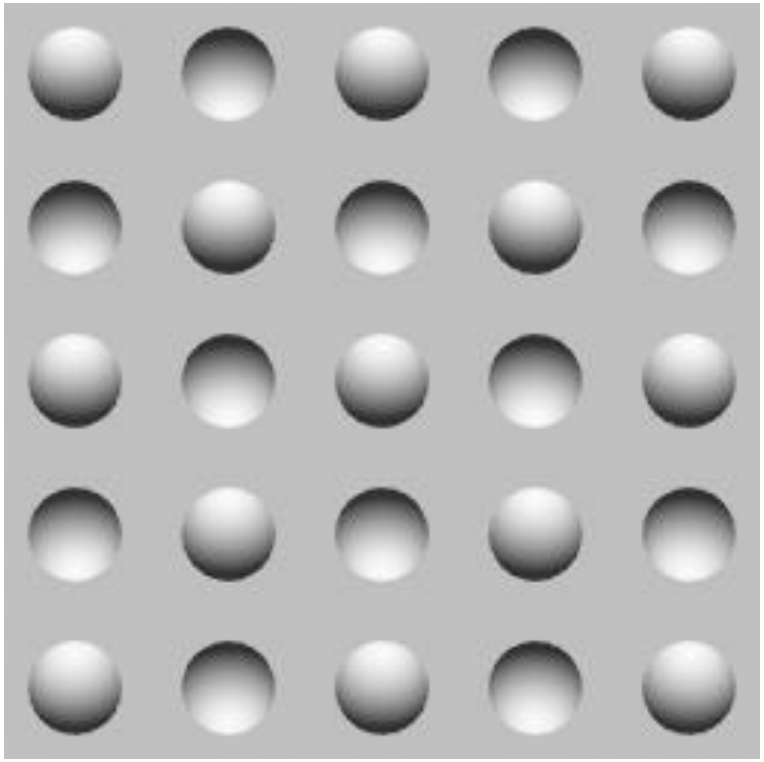
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

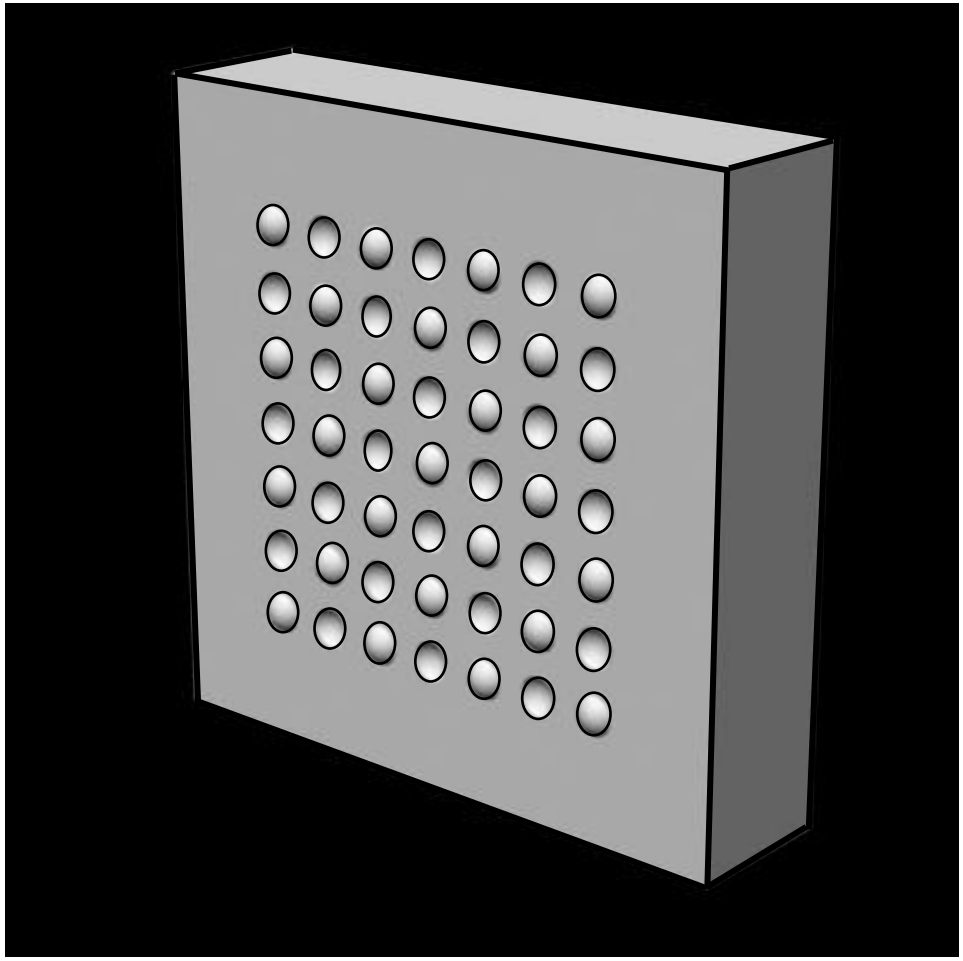
Shape from Shading

Shape from Shading



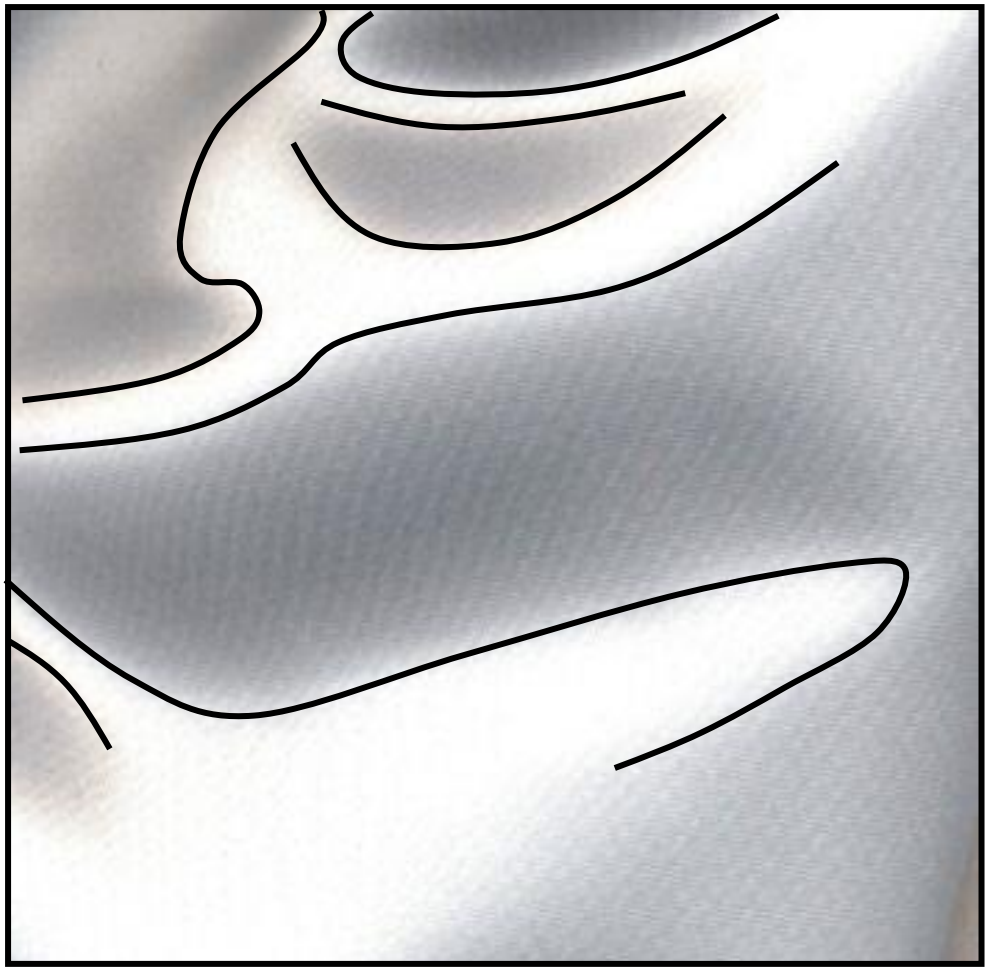
Shape from Shading

Shading is more than contours



Shape from Shading

Shading is more than edge contours



Shape from Shading

Inverting the image formation process

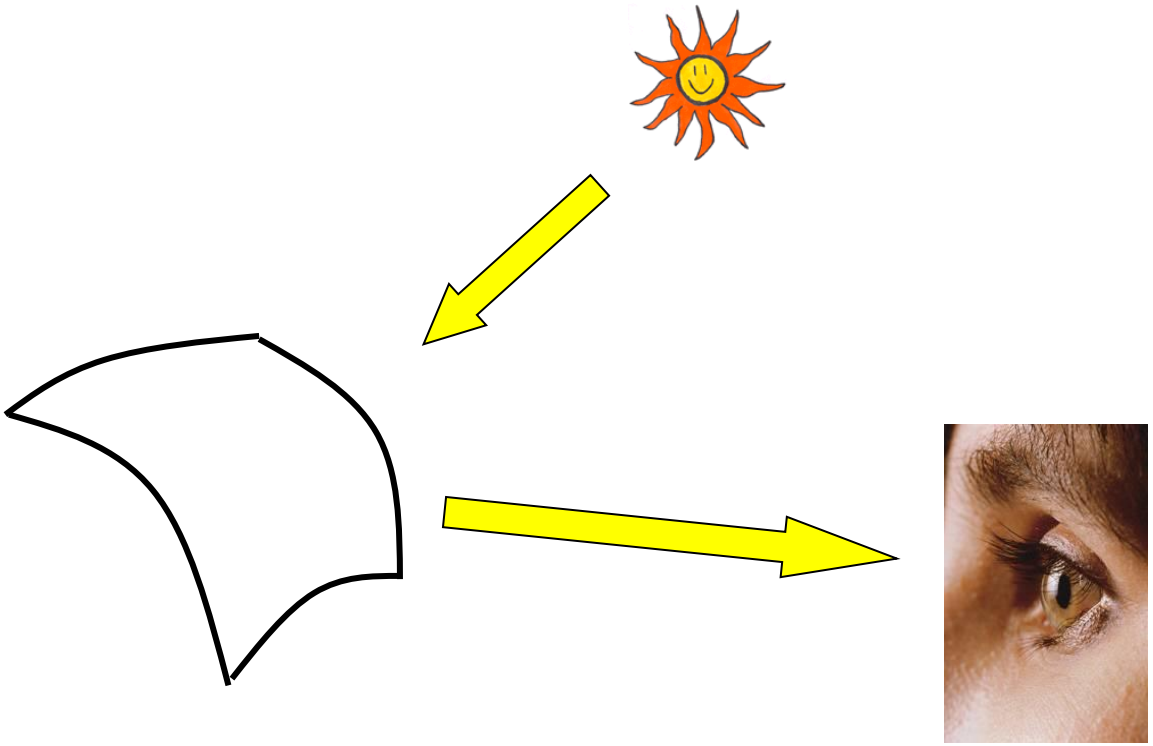


Image formation = “Shading from shape” (and light sources)

Shape from Shading

Image formation

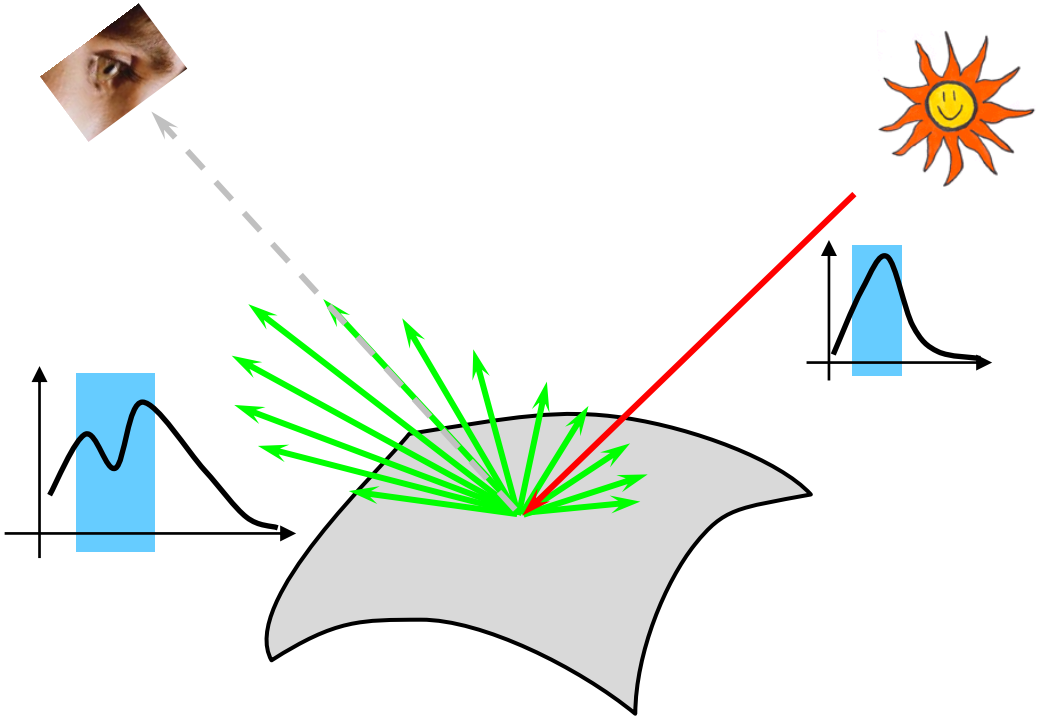
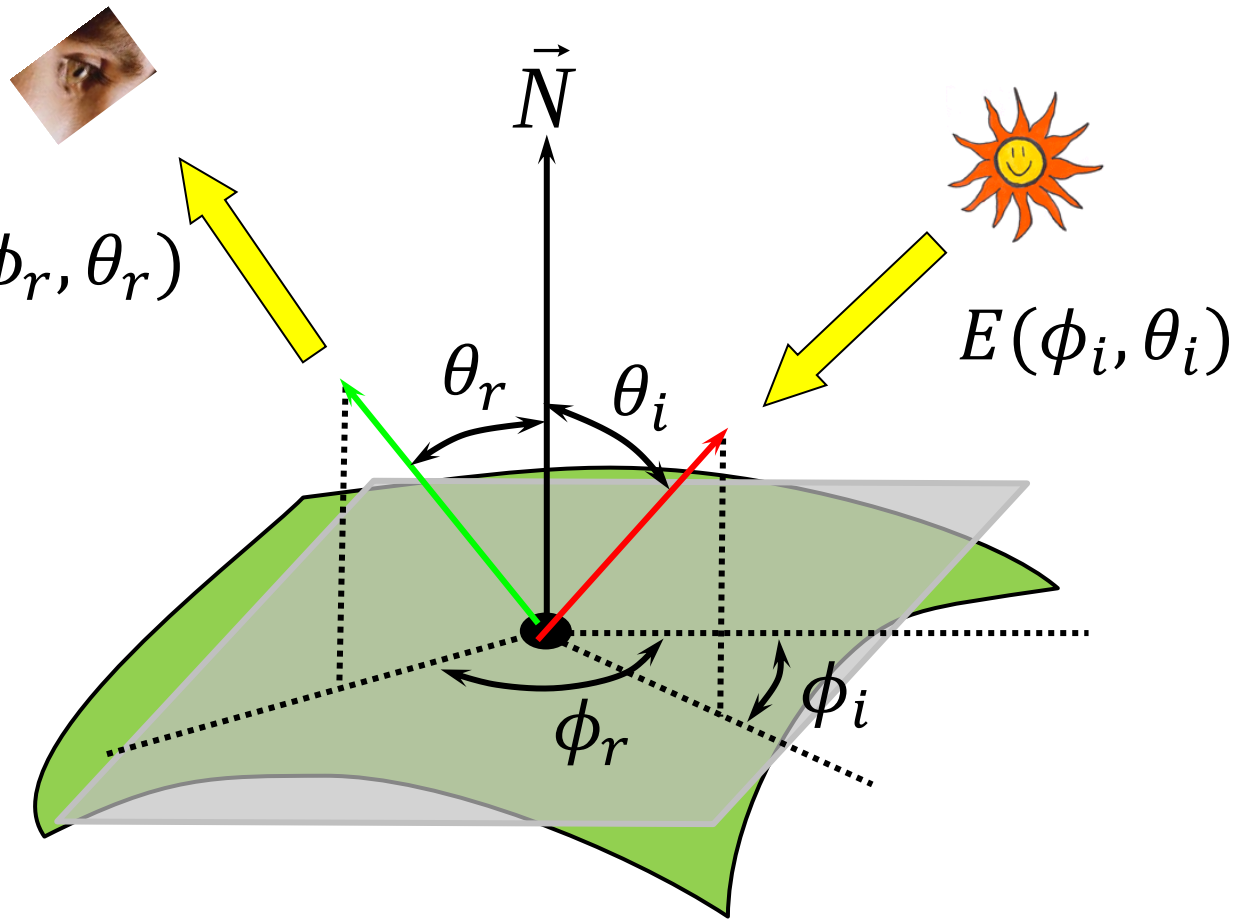


Image formation 1: Where is a world/object point projected in the image plane?

Image formation 2: What is the amount of light that is reflected in the direction of the camera?

Shape from Shading

representation of directions



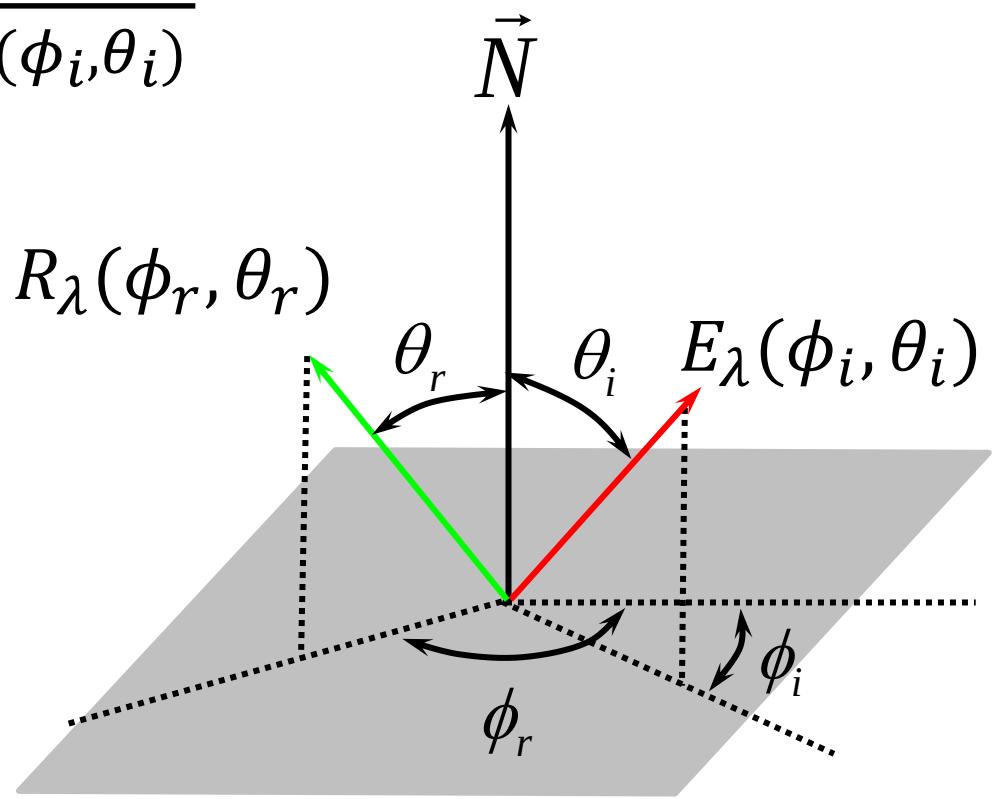
θ - Azimuth angle

ϕ - Zenith angle

Shape from Shading

The Bidirectional Reflectance Distribution Function (BRDF)

$$f_\lambda(\phi_i, \theta_i; \phi_r, \theta_r) = \frac{R_\lambda(\phi_r, \theta_r)}{E_\lambda(\phi_i, \theta_i)}$$



Helmholtz's reciprocity

$$f(\phi_i, \theta_i; \phi_r, \theta_r) = f(\phi_r, \theta_r; \phi_i, \theta_i)$$

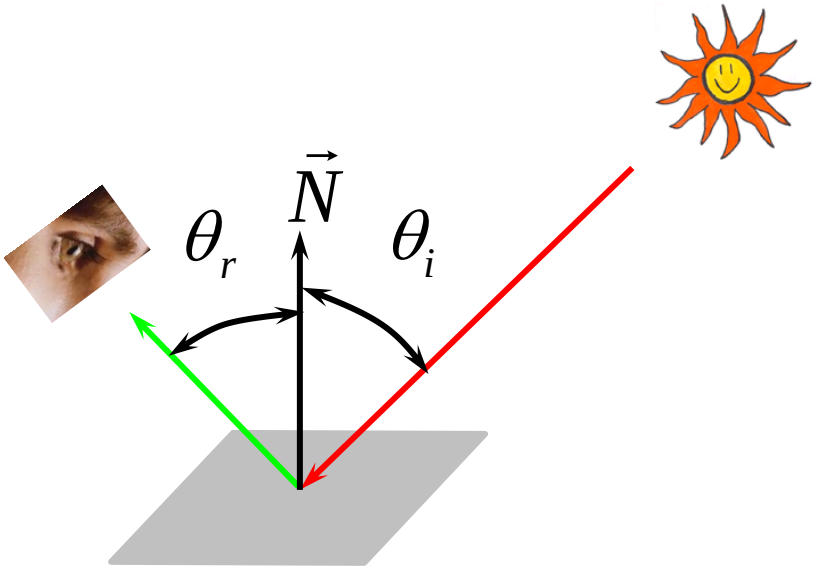
Isotropic materials:

$$f(\phi_i, \theta_i; \phi_r, \theta_r) = f(\phi_i - \phi_r, \theta_i, \theta_r)$$

Shape from Shading

Mirrored (perfectly secular) surfaces

$$f_S(\phi_i, \theta_i; \phi_r, \theta_r) \propto \delta(\theta_r - \theta_i)\delta(\phi_r - \phi_i - \pi)$$

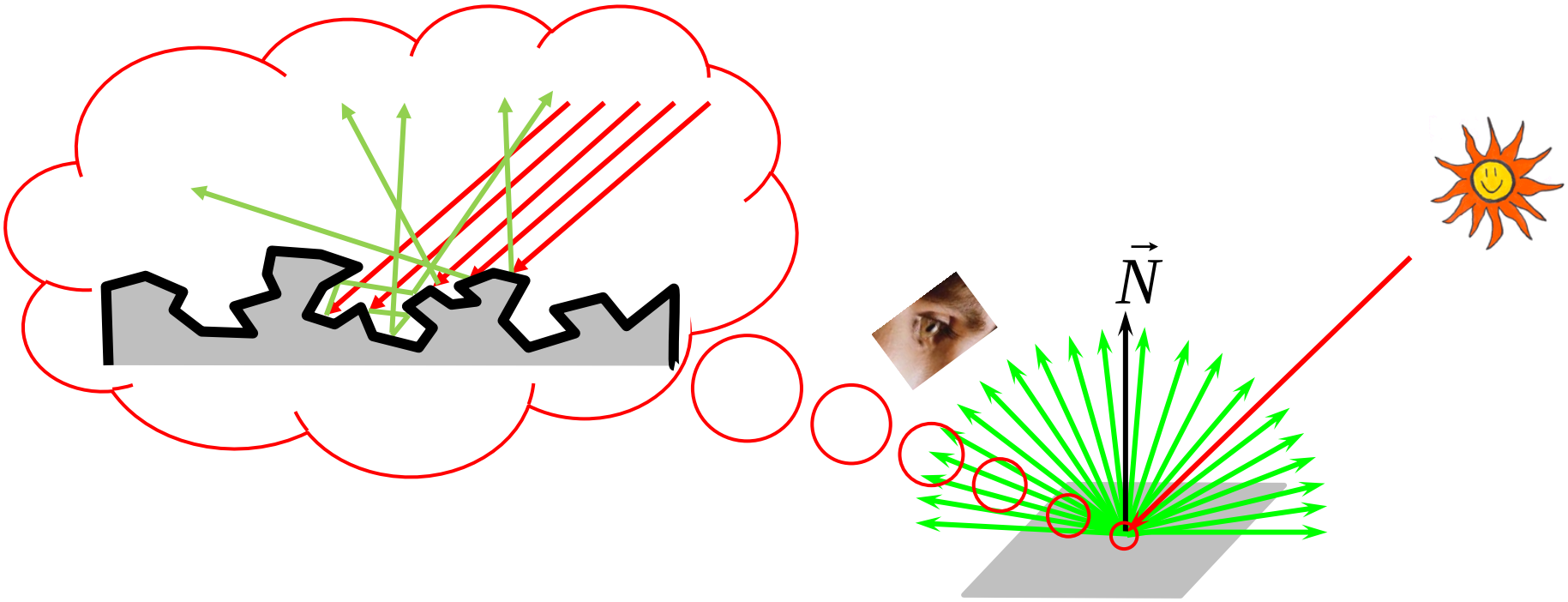


Shape from Shading

Lambertian (perfectly diffused) surfaces

$$f_L(\phi_i, \theta_i; \phi_r, \theta_r) = const \propto \rho$$

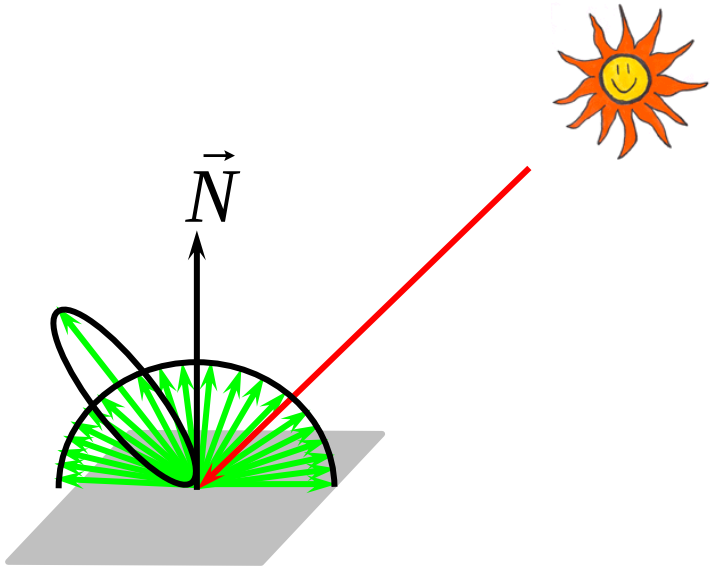
Albedo



Shape from Shading

Realistic (mixed) surfaces

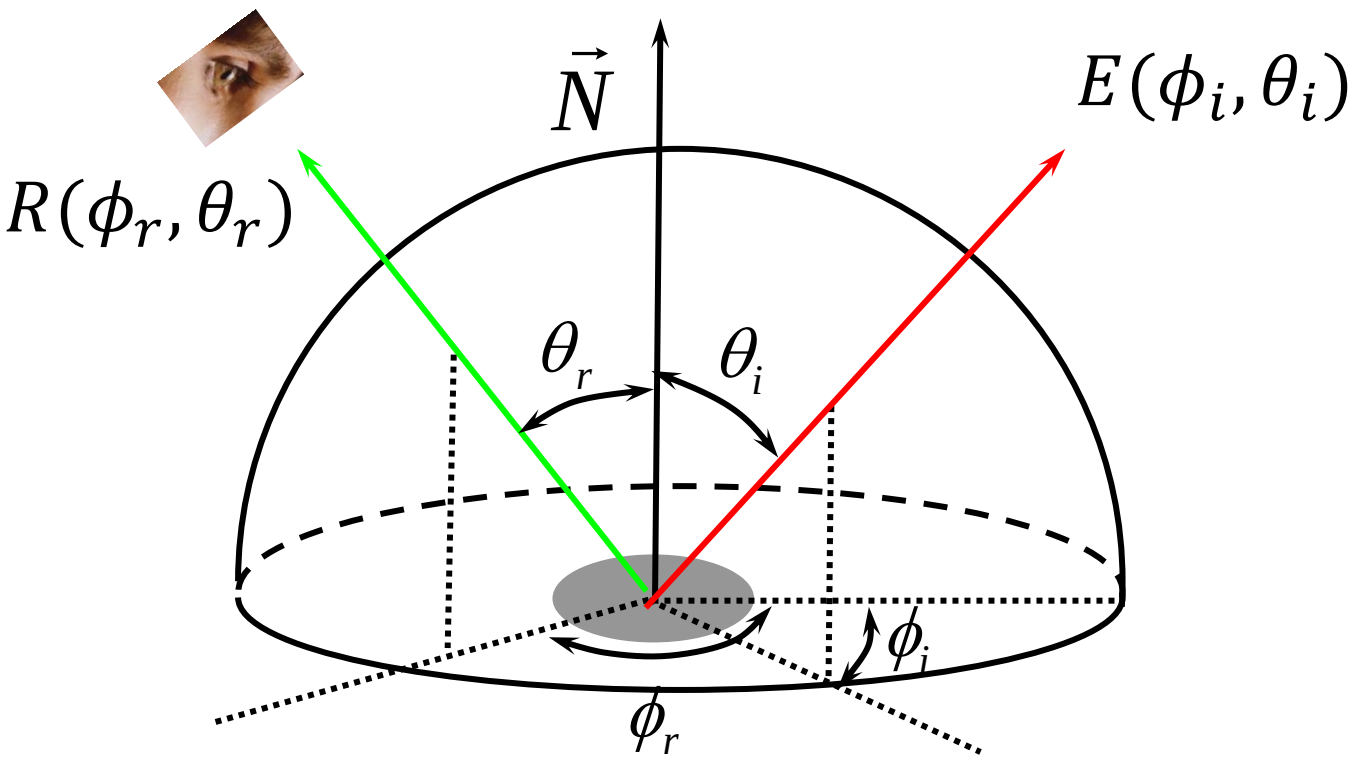
$$f(\phi_i,\theta_i;\phi_r,\theta_r) = \alpha \cdot f_L(\phi_i,\theta_i;\phi_r,\theta_r) + (1-\alpha) f_S(\phi_i,\theta_i;\phi_r,\theta_r)$$



Shape from Shading

What is the intensity reflected in the direction of the camera?

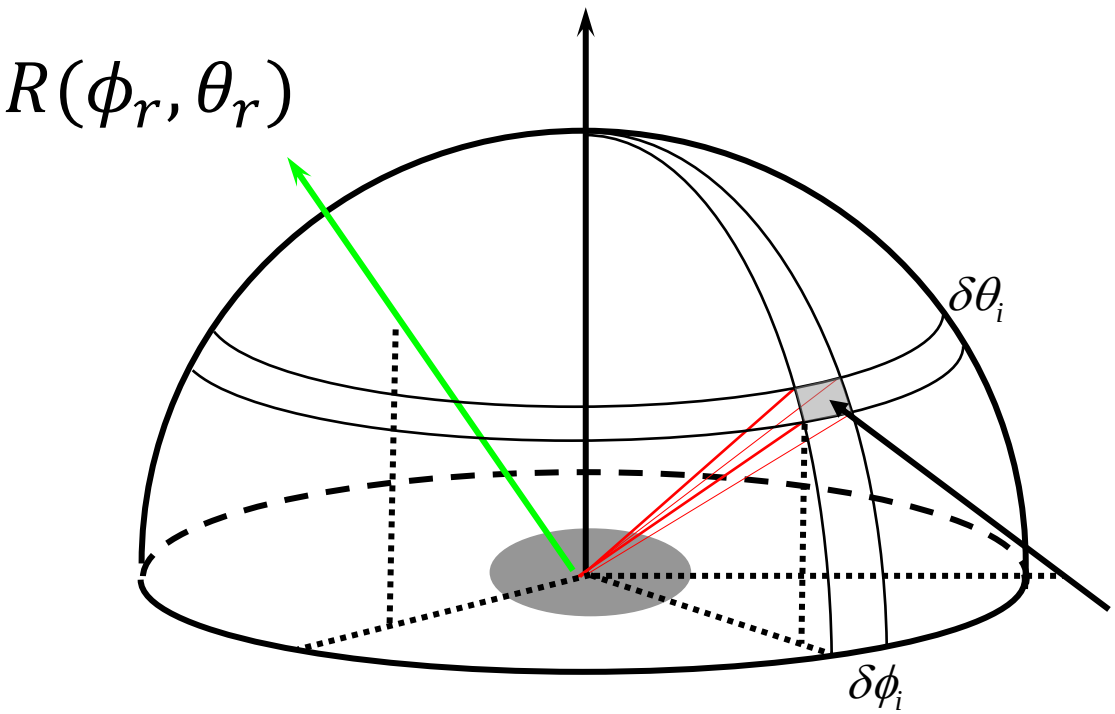
$$f_\lambda(\phi_i, \theta_i; \phi_r, \theta_r) = \frac{R_\lambda(\phi_r, \theta_r)}{E_\lambda(\phi_i, \theta_i)} \Rightarrow R_\lambda(\phi_r, \theta_r) = f_\lambda(\phi_i, \theta_i; \phi_r, \theta_r) E_\lambda(\phi_i, \theta_i)$$

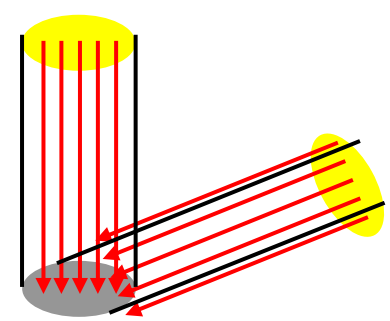


Shape from Shading

Total surface reflection towards the camera

$$R(\phi_r, \theta_r) = \int_{\omega} f(\phi_i, \theta_i; \phi_r, \theta_r) E(\phi_i, \theta_i) \cos\theta_i \, d\omega$$



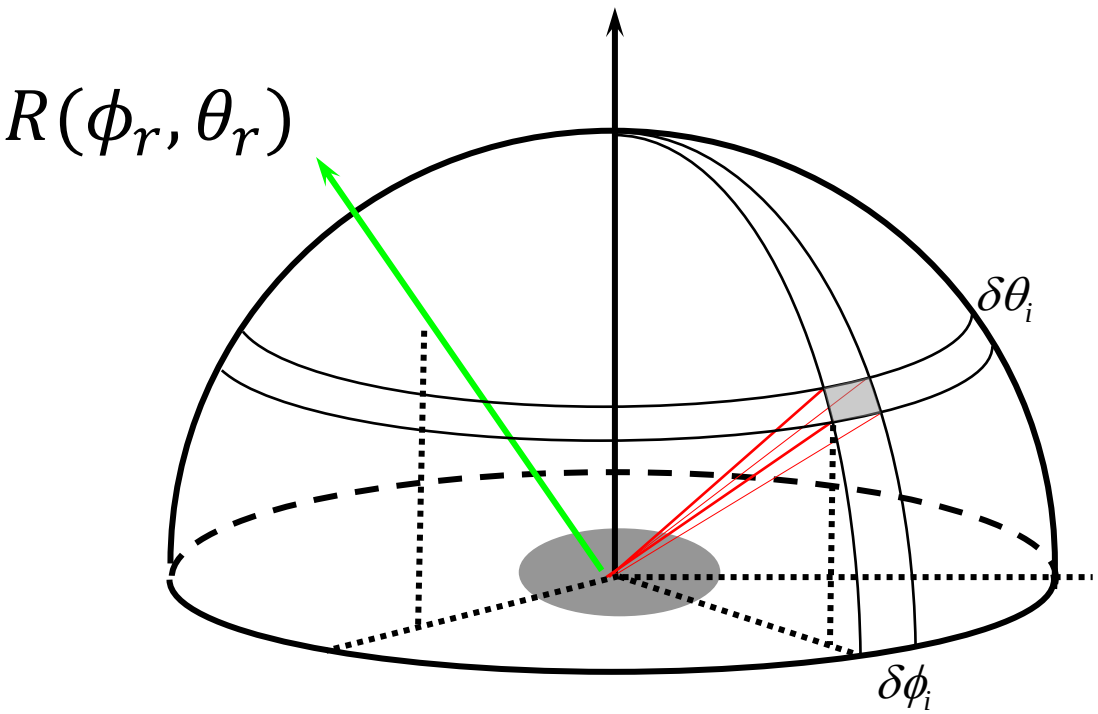


$$\delta\omega = \sin\theta_i \delta\theta_i \delta\phi_i$$

Shape from Shading

Total surface reflection towards the camera

$$R(\phi_r, \theta_r) = \int_{-\pi}^{\pi} \int_{0}^{\pi/2} f(\phi_i, \theta_i; \phi_r, \theta_r) \cdot E(\phi_i, \theta_i) \cdot sin\theta_i \cdot cos\theta_i \delta\theta_i \delta\phi_i$$

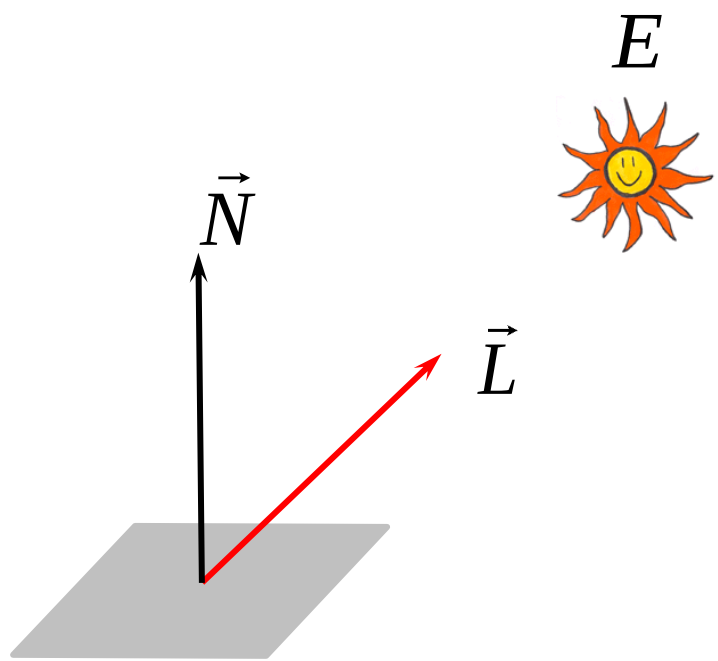


Shape from Shading

Point light source from direction (ϕ_L, θ_L)

$$E(\phi_i, \theta_i) = E \cdot \frac{\delta(\theta_L - \theta_i) \cdot \delta(\phi_L - \phi_i)}{\sin\theta_L}$$

$$\int_{-\pi}^{\pi} \int_{0}^{\pi/2} E(\phi_i, \theta_i) \cdot \sin\theta_i \cdot \delta\theta_i \delta\phi_i = E$$



Shape from Shading

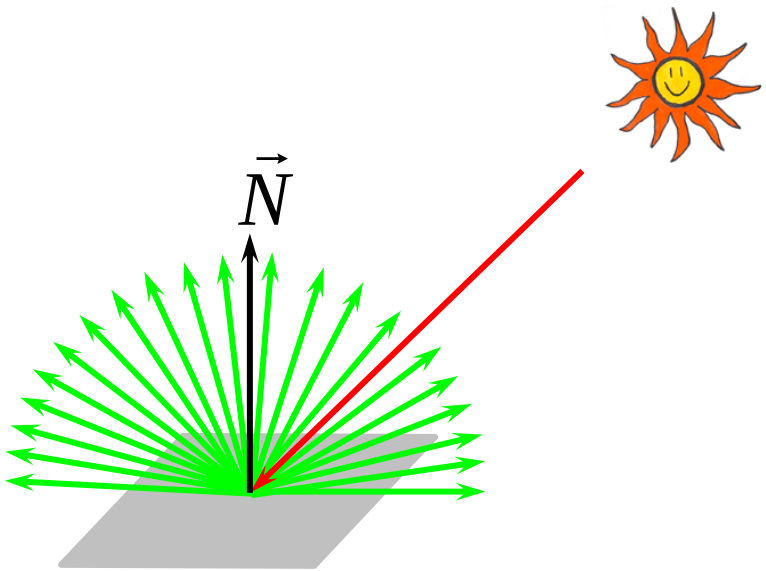
Lambertian (perfectly diffused) surfaces

$$f_L(\phi_i, \theta_i; \phi_r, \theta_i) = const = \bar{f} = \frac{1}{\pi}\rho$$

Albedo

$$\int_{-\pi}^{\pi}\int_{0}^{\pi/2} \overline{f} \cdot \sin\theta_r \cdot \cos\theta_r \cdot \delta\theta_r \delta\phi_r = 1$$

$$\pi \overline{f} = 1$$



Shape from Shading

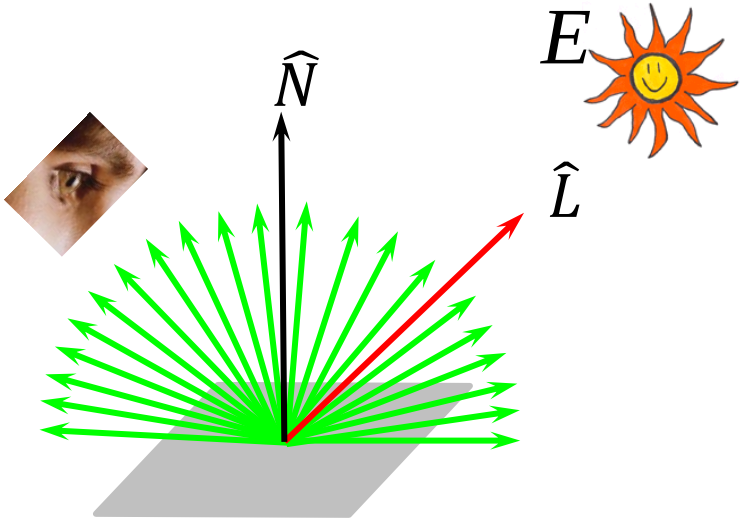
Surface brightness – appearance in the Lambertian case and point light source

$$f_L(\phi_i,\theta_i;\phi_r,\theta_r) = \rho\frac{1}{\pi}$$

$$E(\phi_i,\theta_i) = \frac{\delta(\theta_L-\theta_i)\delta(\phi_L-\phi_i)}{\sin\theta_L}$$

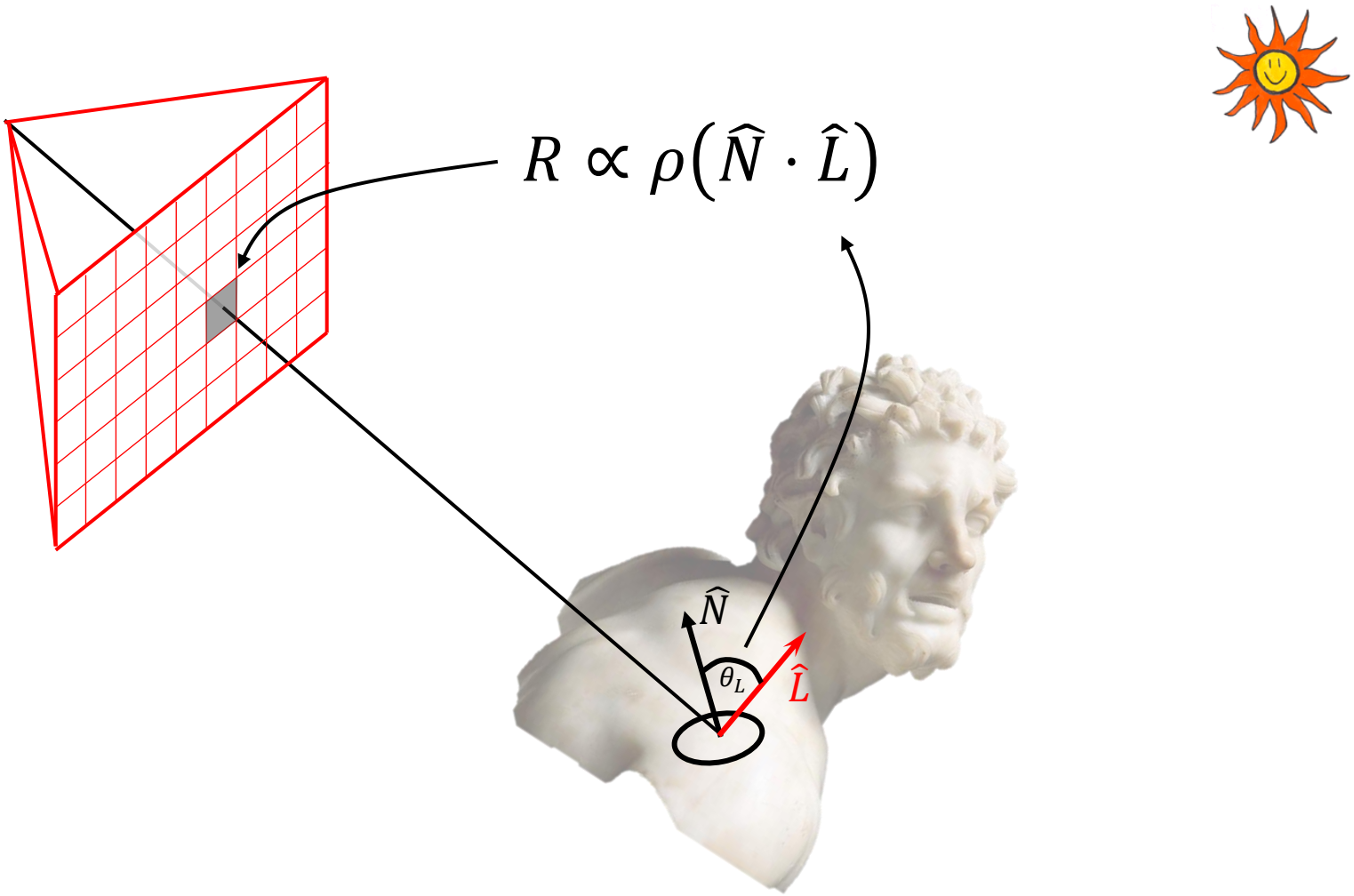
$$I(x,y) \propto R(\phi_r,\theta_r) = \int_{-\pi}^{\pi}\int_{0}^{\pi/2} f(\phi_i,\theta_i;\phi_r,\theta_r)\cdot E_\lambda(\phi_i,\theta_i)\cdot sin\theta_i\cdot cos\theta_i\delta\theta_i\delta\phi_i$$

$$R = \rho\frac{1}{\pi}\cdot E\cdot cos\theta_L \propto \rho\left(\hat{N}\cdot\hat{L}\right)$$



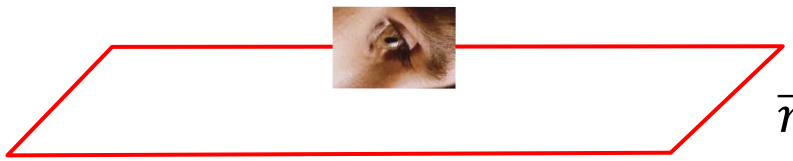
Shape from Shading

So… How can we use all this for shape from shading?



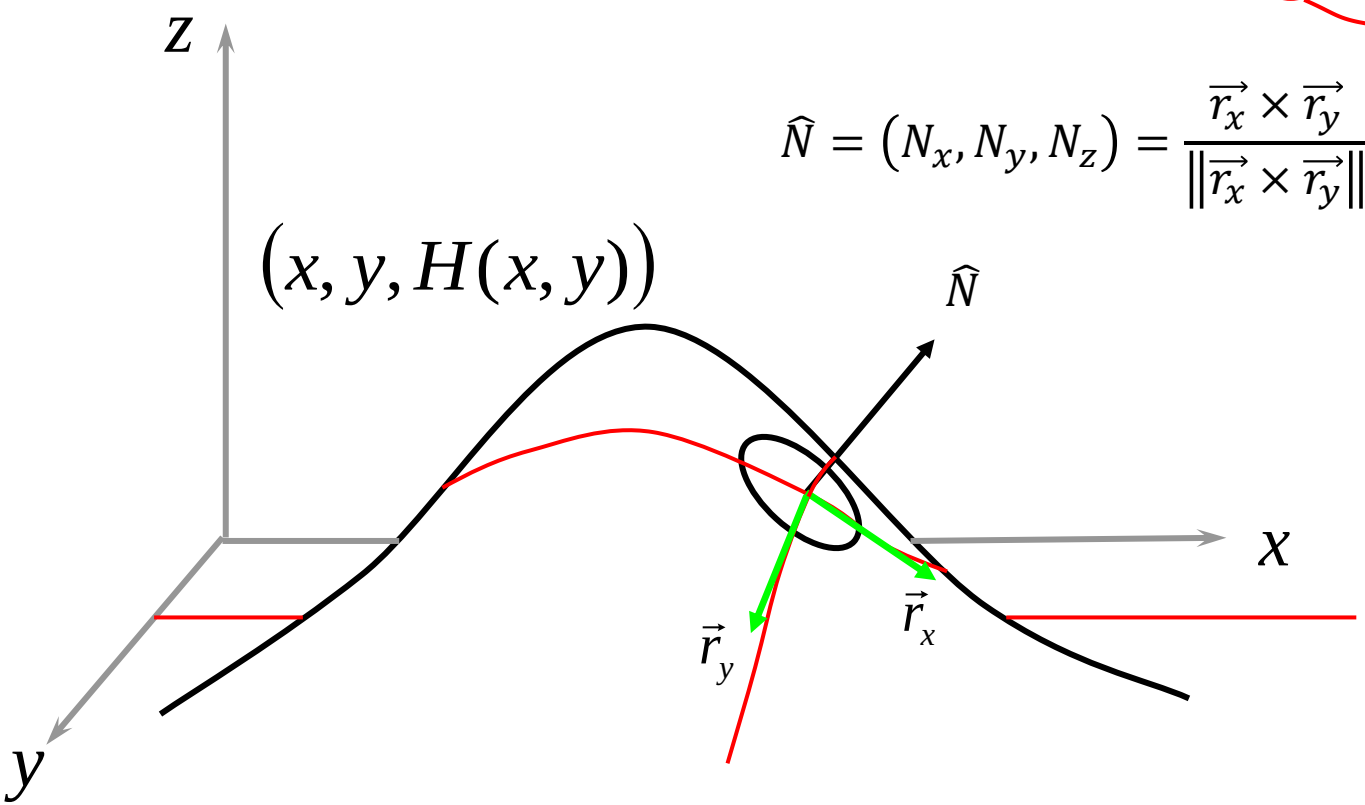
Shape from Shading

Shape description via tangent plane and normal surface of the object at each point



$$\vec{r_x} = \left(1, 0, \frac{\partial H}{\partial x}\right) \qquad \vec{r_y} = \left(1, 0, \frac{\partial H}{\partial y}\right)$$

$$\hat{N} = (N_x, N_y, N_z) = \frac{\vec{r_x} \times \vec{r_y}}{\|\vec{r_x} \times \vec{r_y}\|} = \frac{(-p, -q, 1)}{\sqrt{p^2 + q^2 + 1}}$$



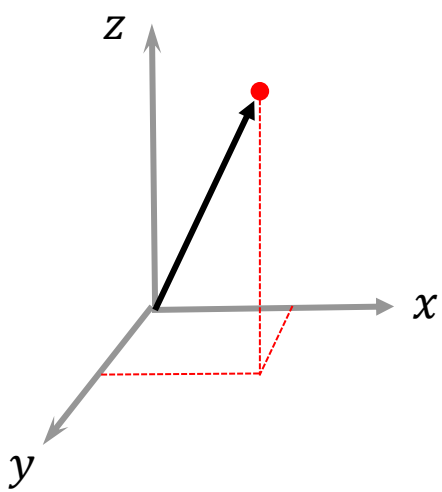
Shape from Shading

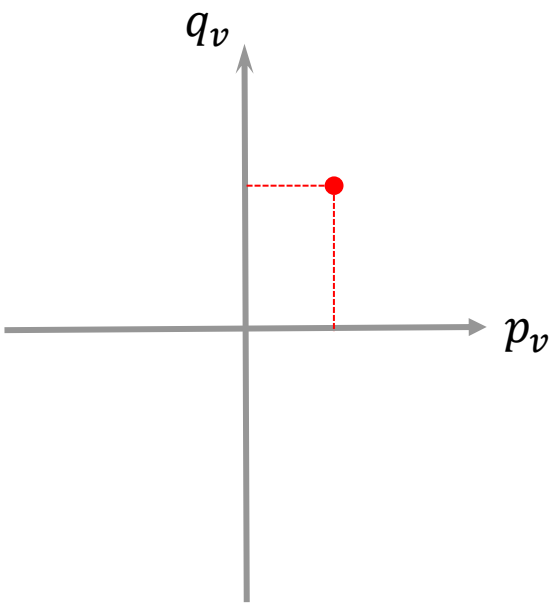
Different ways to represent directions (and function thereof)

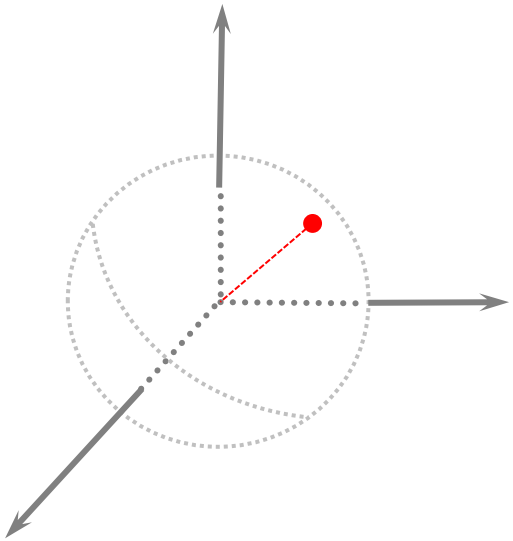
$$V = \frac{(v_x, v_y, v_z)}{\sqrt{v_x^2 + v_y^2 + v_z^2}}$$

$$V = \frac{(-p_v, -q_v, 1)}{\sqrt{p_v^2 + q_v^2 + 1}}$$

$$V \in S^2$$







$$f(V) = f(v_x, v_y, v_z)$$

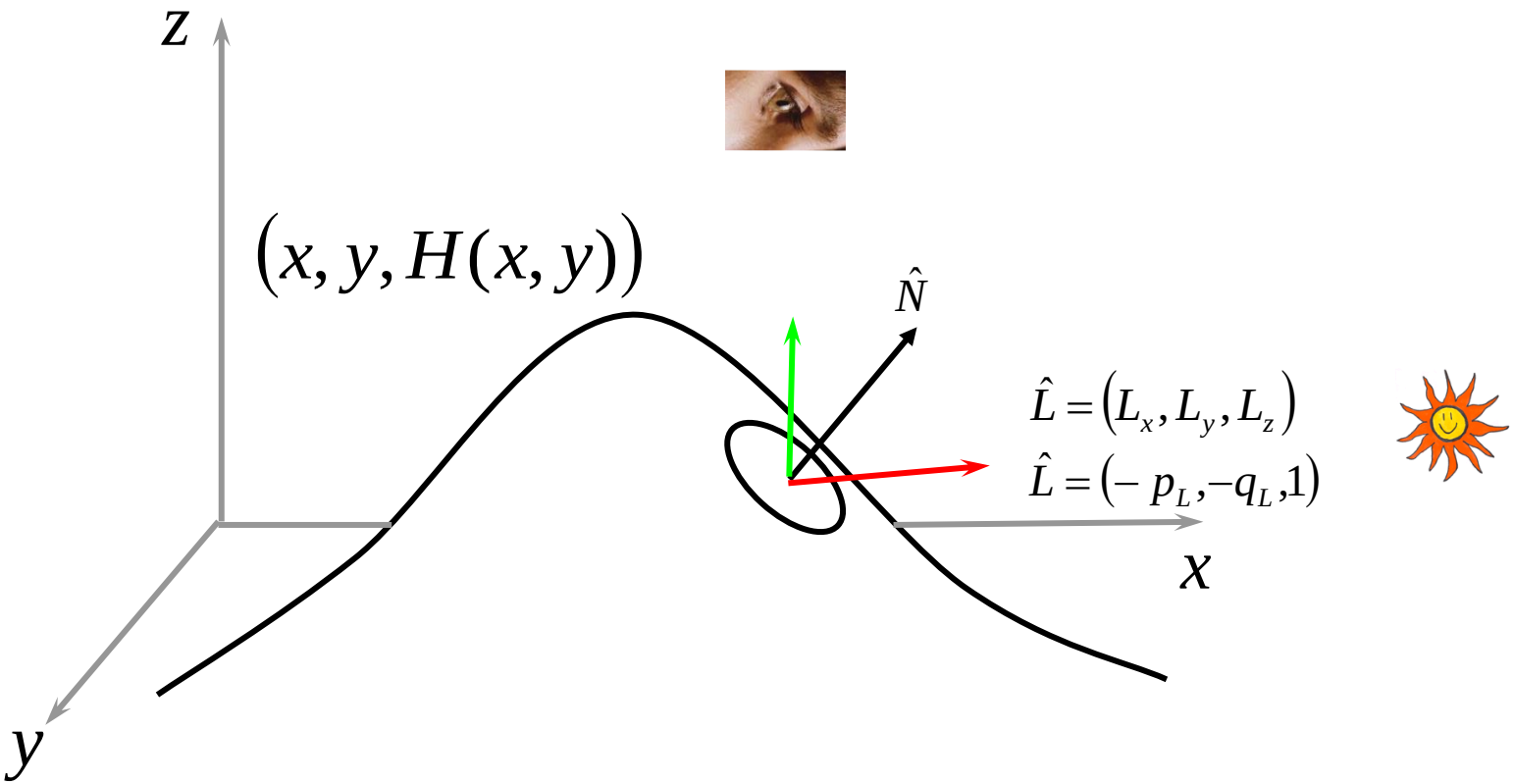
$$f(V) = f(p_v, q_v)$$

$$f(V)$$

Shape from Shading

Shading on Lambertian surface – General point source

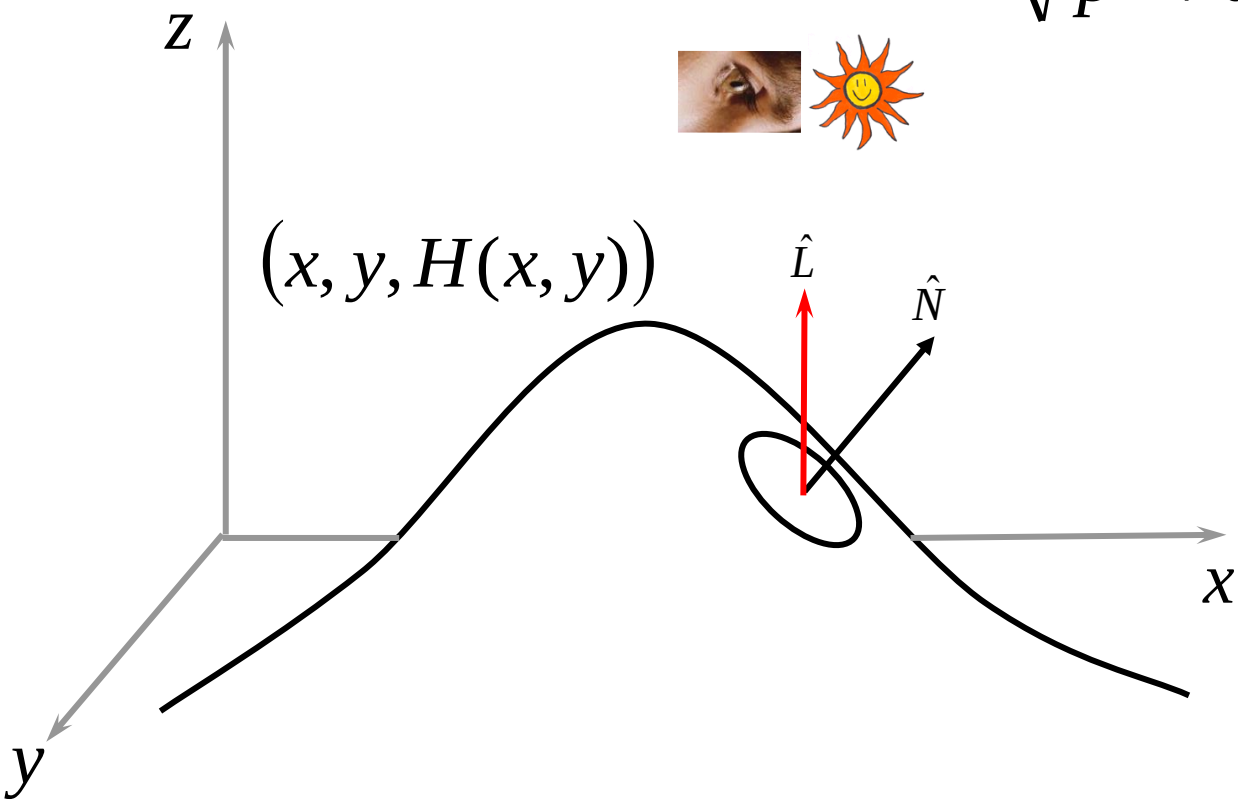
$$I = \rho(\hat{N}\cdot\hat{L}) = \rho\frac{-p\cdot L_x - q\cdot L_y + L_z}{\sqrt{p^2+q^2+1}\sqrt{L_x{}^2+L_y{}^2+L_z{}^2}} = \rho\frac{p\cdot p_L + q\cdot q_L + 1}{\sqrt{p^2+q^2+1}\sqrt{p_L{}^2+q_L{}^2+1}}$$



Shape from Shading

Shading on Lambertian surface – Overhead point source

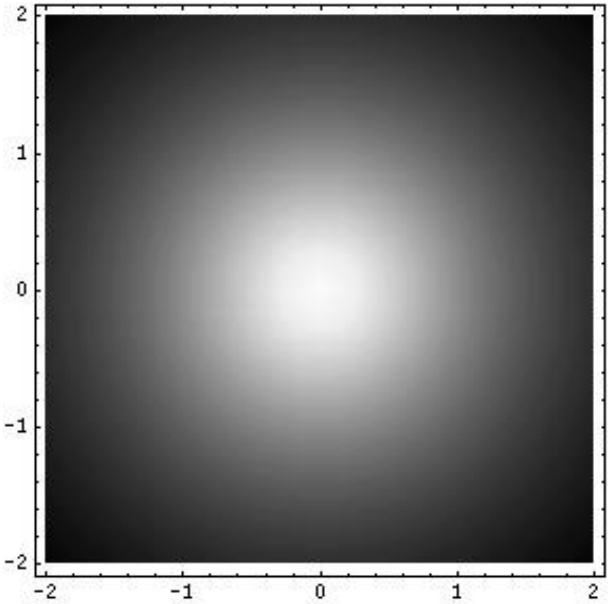
$$I(x,y) = \rho(\hat{N} \cdot [0,0,1]) = \rho \frac{1}{\sqrt{p^2 + q^2 + 1}} = R(p,q)$$

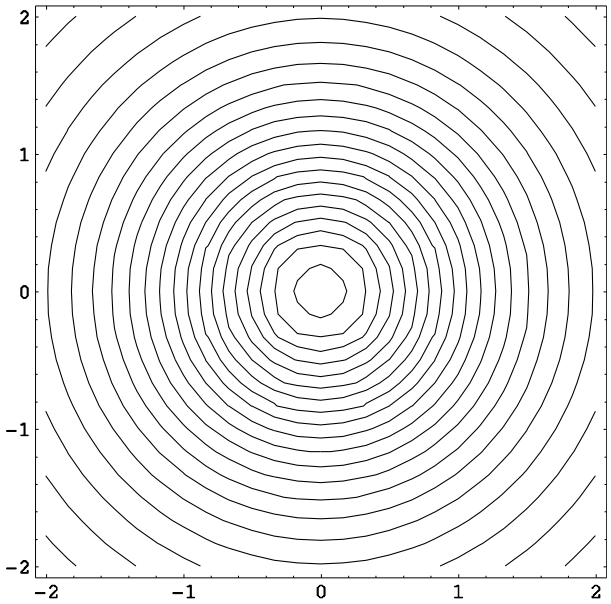


Shape from Shading

The Reflectance Map – Lambertian surface from overhead source position

$$R(p,q) = \frac{1}{\sqrt{p^2 + q^2 + 1}}$$

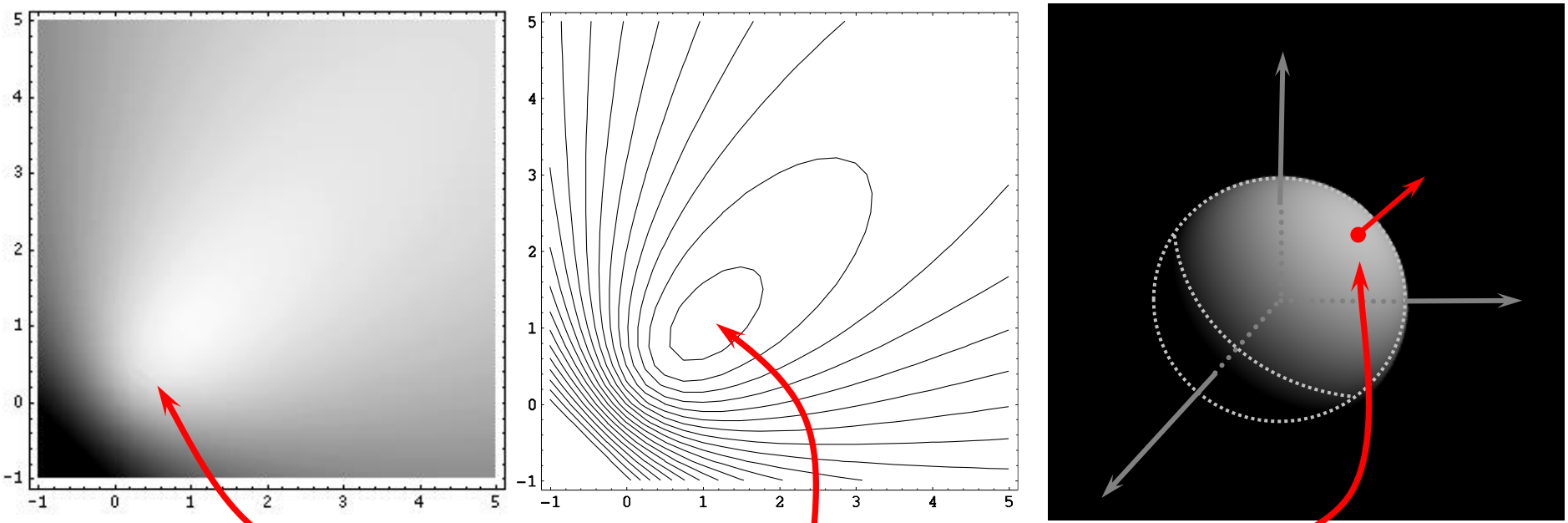




Shape from Shading

The Reflectance Map – Lambertian surface from general source position

$$R(p,q) = \frac{p \cdot p_L + q \cdot q_L + 1}{\sqrt{p^2 + q^2 + 1}\sqrt{{p_L}^2 + {q_L}^2 + 1}}$$

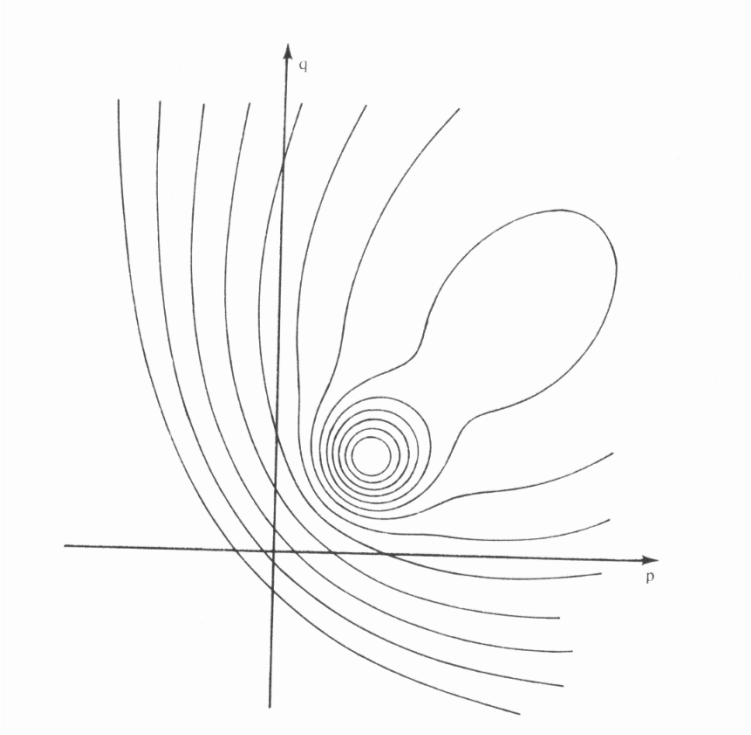


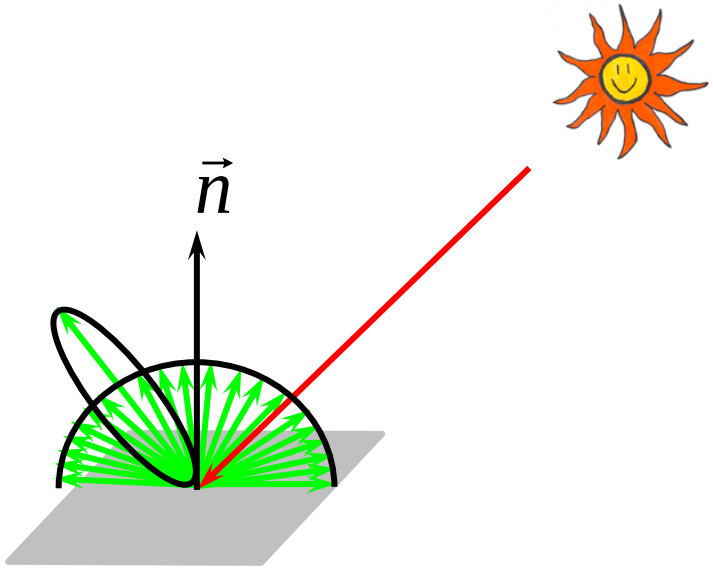
Gradient point of maximum brightness

Shape from Shading

The Reflectance Map – typical real surfaces

$$R(p,q)$$

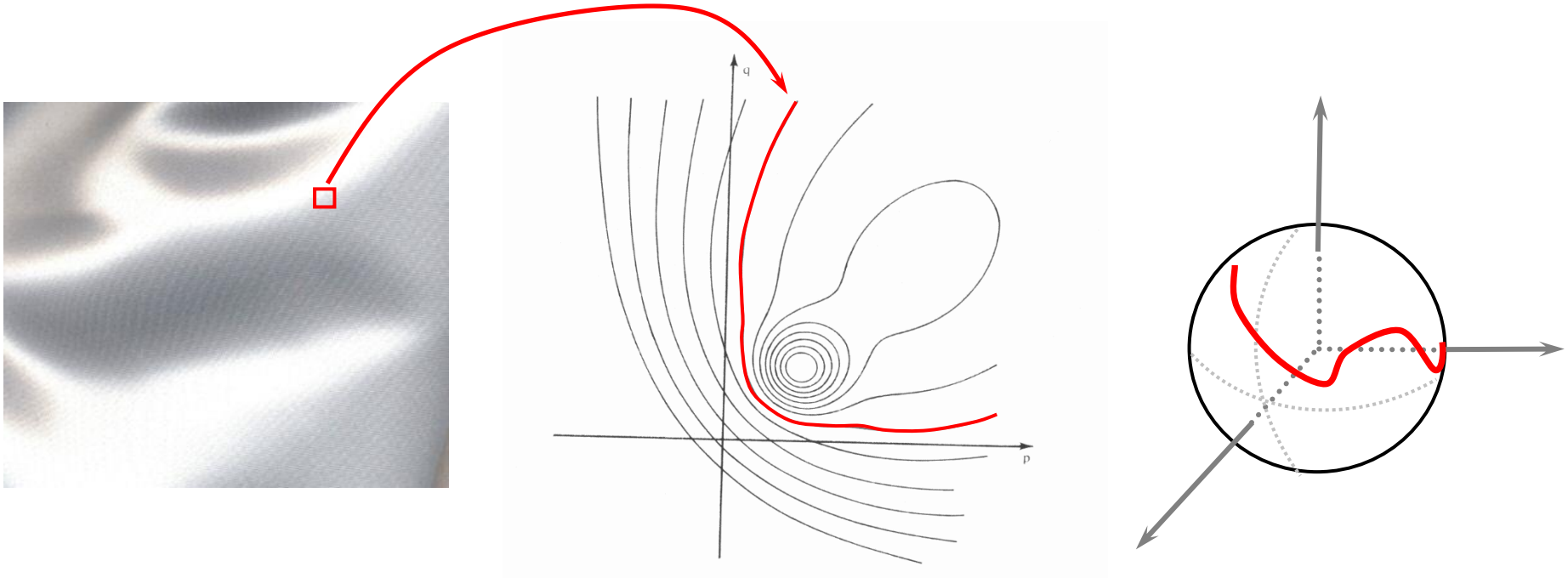




Shape from Shading

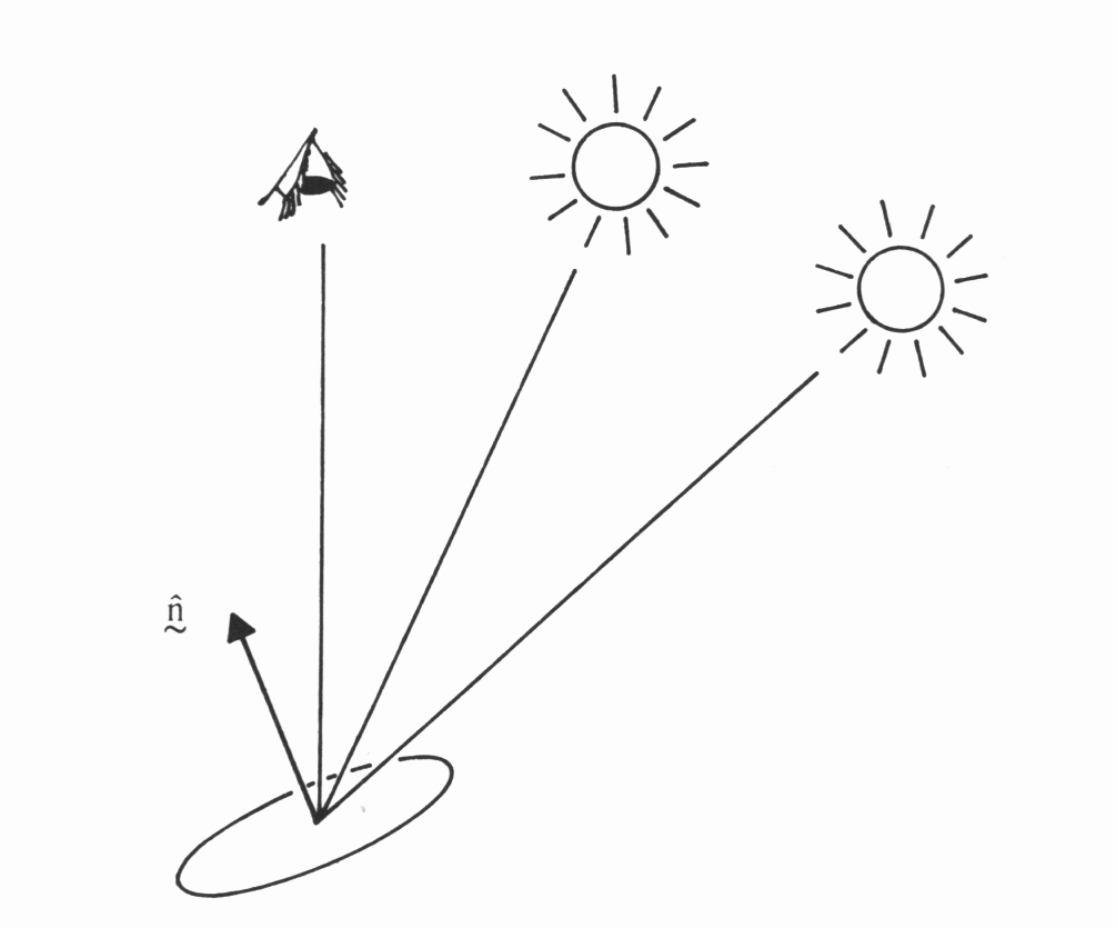
Surface orientation from shading

$$I(x, y) = R(p, q)$$



Shape from Shading

Photometric stereo

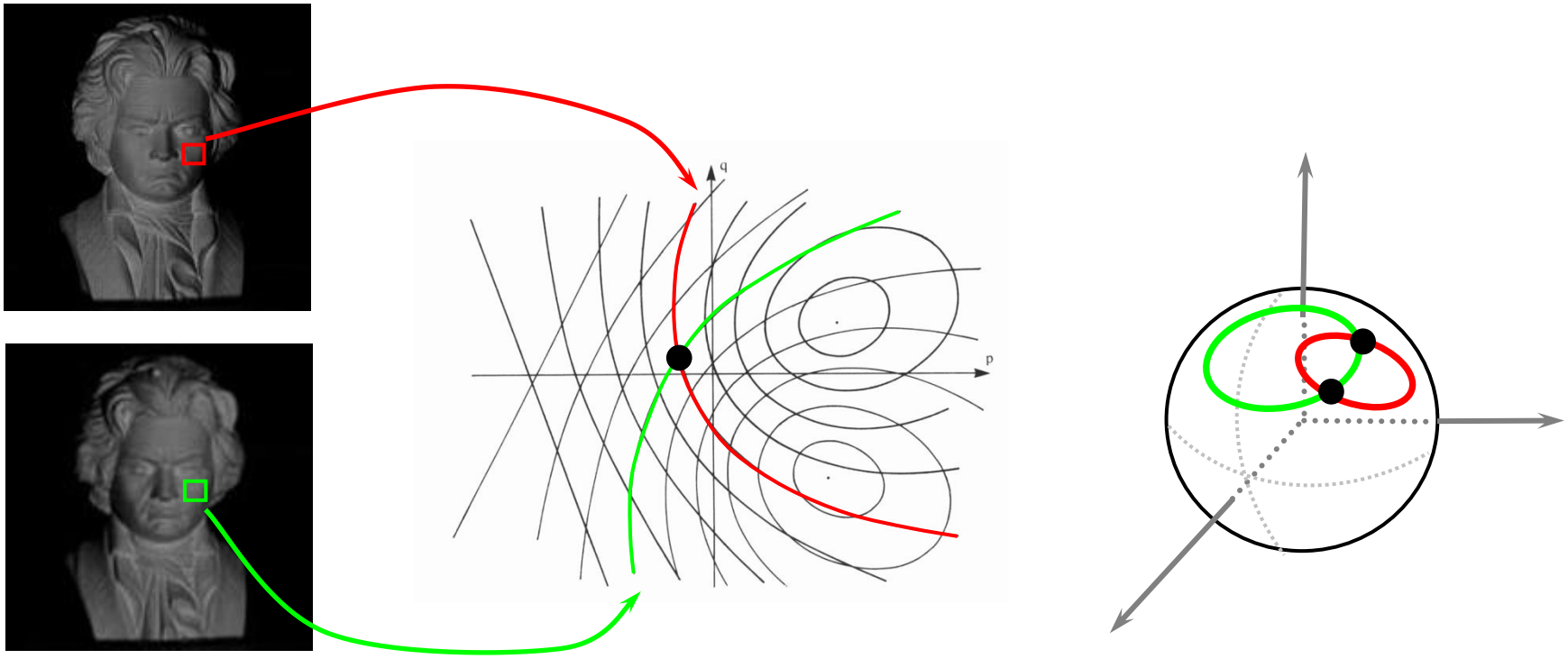


Shape from Shading

Photometric stereo

$$I_1(x, y) = R_1(p, q)$$

$$I_2(x, y) = R_2(p, q)$$



Shape from Shading

The SFS problem (special case)

Given *I(x,y)* of an (orthographic) projection of *H(x,y)*, and the reflectance map *R(p,q)*, find *H(x,y)* everywhere.

$$I(x,y) = R(p,q) = R\left(\frac{\partial}{\partial x}H(x,y), \frac{\partial}{\partial y}H(x,y)\right)$$

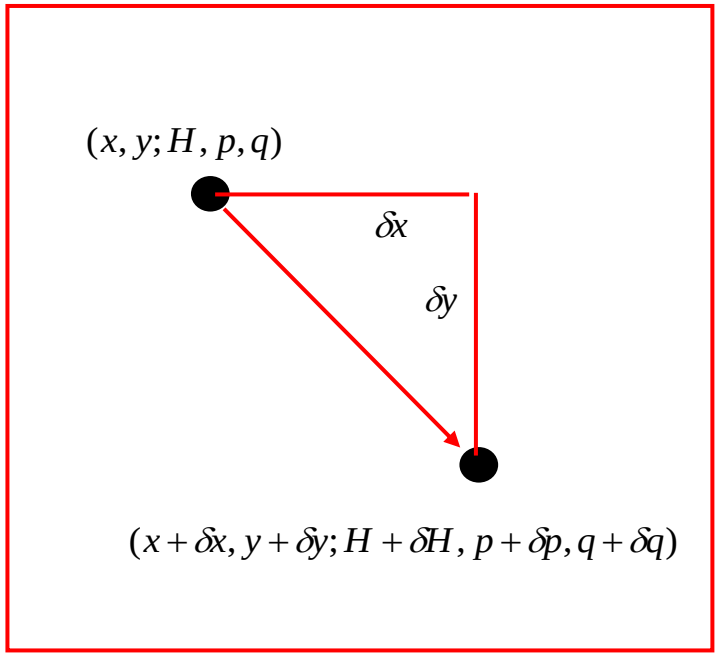
Shape from Shading

Shape recovery via characteristic strips

$$I(x,y) = R(p(x,y), q(x,y))$$

$$p(x,y) = \frac{\partial}{\partial x} H(x,y)$$

$$q(x,y) = \frac{\partial}{\partial y} H(x,y)$$



$$H(x+\delta x, y+\delta y) \cong H(x,y) + p\,\delta x + q\,\delta y$$

$$p(x+\delta x, y+\delta y) \cong p(x,y) + \frac{\partial p}{\partial x}\delta x + \frac{\partial p}{\partial y}\delta y$$

$$q(x+\delta x, y+\delta y) \cong q(x,y) + \frac{\partial q}{\partial x}\delta x + \frac{\partial q}{\partial y}\delta y$$

Shape from Shading

Shape recovery via characteristic strips

$$I(x,y) = R(p(x,y), q(x,y))$$

$$\frac{\partial}{\partial x} I(x,y) = \frac{\partial R(p,q)}{\partial p} \overset{?}{\boxed{\frac{\partial p(x,y)}{\partial x}}} + \frac{\partial R(p,q)}{\partial q} \overset{?}{\boxed{\frac{\partial q(x,y)}{\partial x}}}$$

$$\frac{\partial}{\partial y} I(x,y) = \frac{\partial R(p,q)}{\partial p} \boxed{\frac{\partial p(x,y)}{\partial y}} + \frac{\partial R(p,q)}{\partial q} \boxed{\frac{\partial q(x,y)}{\partial y}}$$

Shape from Shading

Shape recovery via characteristic strips

$$I(x,y) = R(p(x,y), q(x,y))$$

$$\frac{\partial}{\partial x} I(x,y) = \frac{\partial R(p,q)}{\partial p}\frac{\partial p(x,y)}{\partial x} + \frac{\partial R(p,q)}{\partial q}\frac{\partial p(x,y)}{\partial y}$$

$$\frac{\partial}{\partial y} I(x,y) = \frac{\partial R(p,q)}{\partial p}\frac{\partial q(x,y)}{\partial x} + \frac{\partial R(p,q)}{\partial q}\frac{\partial q(x,y)}{\partial y}$$

Shape from Shading

Shape recovery via characteristic strips

$$I(x,y) = R(p(x,y), q(x,y))$$

$$\delta H \cong p\,\delta x + q\,\delta y$$

$$\frac{\partial}{\partial x} I(x,y) = \nabla R \cdot \nabla p$$

$$\delta p \cong \frac{\partial p}{\partial x}\delta x + \frac{\partial p}{\partial y}\delta y = \nabla p \cdot (\delta x, \delta y)$$

$$\frac{\partial}{\partial y} I(x,y) = \nabla R \cdot \nabla q$$

$$\delta q \cong \frac{\partial q}{\partial x}\delta x + \frac{\partial q}{\partial y}\delta y = \nabla q \cdot (\delta x, \delta y)$$

A smart choice

$$\delta x = \frac{\partial R(p,q)}{\partial p}\delta s$$

$$\delta y = \frac{\partial R(p,q)}{\partial q}\delta s$$

$$\Rightarrow$$

$$\delta p \cong \frac{\partial}{\partial x} I(x,y) \cdot \delta s$$

$$\delta q \cong \frac{\partial}{\partial y} I(x,y) \cdot \delta s$$

Shape from Shading

Shape recovery via characteristic strips

$$\delta x = R_p \delta s \qquad \dot{x} = R_p$$

$$\delta y = R_q \delta s \qquad \dot{y} = R_q$$

$$\delta H = \left(pR_p + qR_q\right)\delta s \quad \Longrightarrow \quad \dot{H} = pR_p + qR_q$$

$$\delta p = I_x \delta s \qquad \dot{p} = I_x$$

$$\delta q = I_y \delta s \qquad \dot{q} = I_y$$

Shape from Shading

Shape recovery via characteristic strips

Shape from Shading via Characteristic Curves

Given

- *I(x,y)* of an (orthographic) projection of unknown *H(x,y)*
- The reflectance map *R(p,q)*
- Initial data x_0, y_0, $H(x_0,y_0)$, $p(x_0,y_0)$, $q(x_0,y_0)$

Develop a curve solution on *H(x,y)* by taking small steps of size δs via the system

$$\delta x = R_p \delta s$$

$$\delta y = R_q \delta s$$

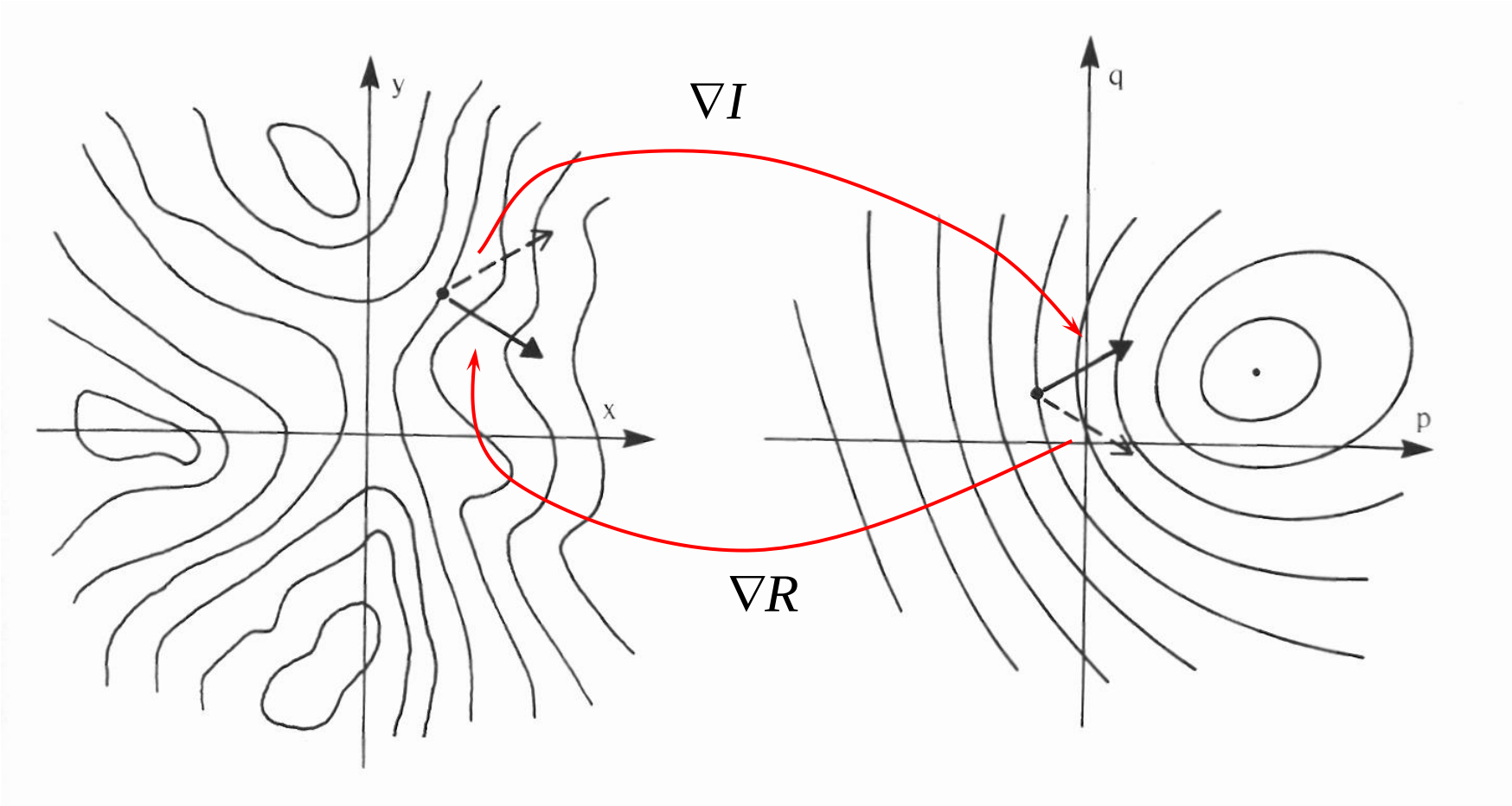
$$\delta H = \left(pR_p + qR_q\right)\delta s$$

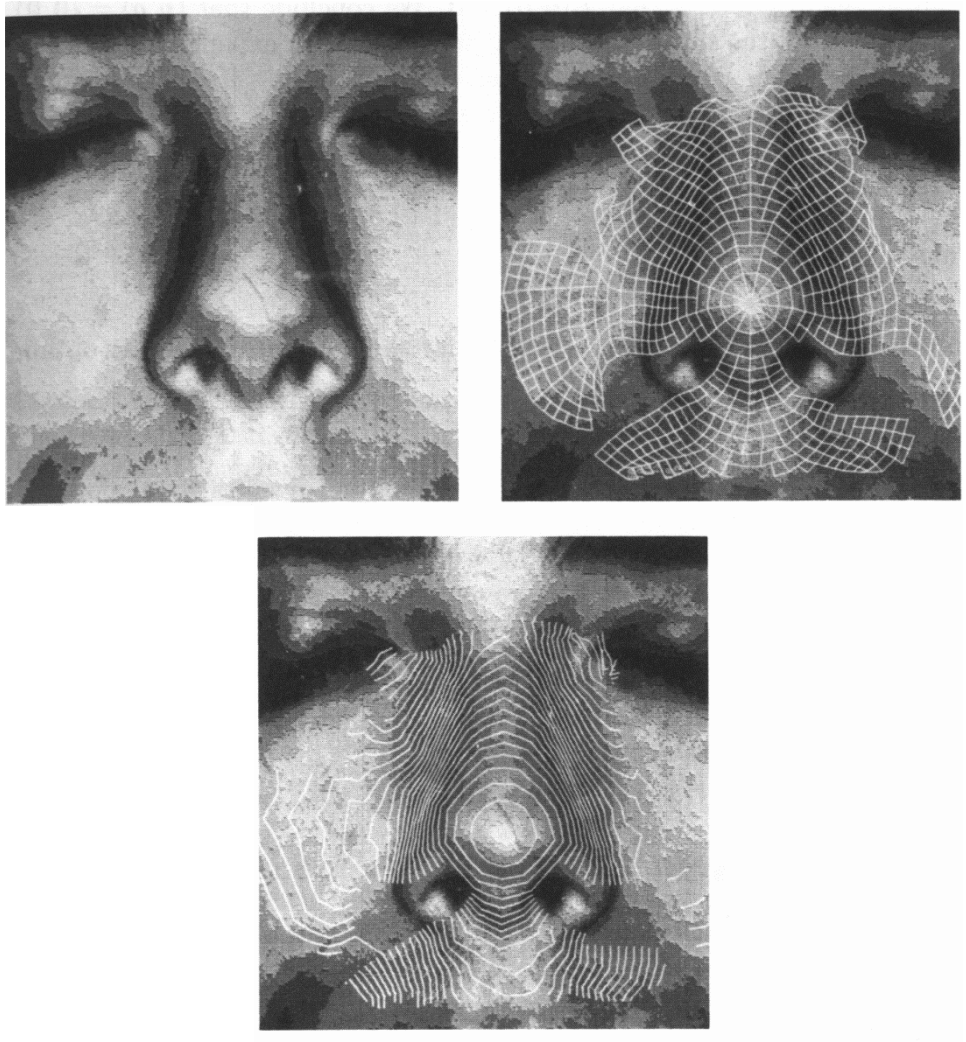
$$\delta p = I_x \delta s$$

$$\delta q = I_y \delta s$$

Shape from Shading

Shape recovery via characteristic strips



Shape from Shading

Shape recovery via characteristic strips